

Dadmehr Daniel Ghasemfar

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TECHNICAL SKILLS

Languages: Java, Python, C++, MATLAB, Maple, TensorFlow, HTML/CSS/JavaScript, ROS
Software: CAD [SolidWorks, Fusion, OnShape], FEA and CFD [Ansys, SolidWorks, In-House], PCB [Eagle, KiCad]
Equipment: 3D Printing [SDF, SLS, Metal], Machining [Mill, CNC, Lathe, Woodwork], Lasercutting, Soldering
Theoretical: solving PDE's and ODE's, Numerical Analysis, Accurate GPU Simulations, PID's and Advanced Controls
Soft Skills: Project Lead, Clear Communication, Professional Documentation, Fluent Farsi, Conversational Spanish

EDUCATION

Duke University

Durham, NC

Bachelor of Science in Mechanical Engineering with Minors in Math and Psychology

Aug 2020 – Dec 2024

- Engineering GPA: 3.6 → Math GPA: 3.5
- Classes: intro/intermediate robotics, intermediate dynamics, chaos and perturbation theory, control theory, heat and mass transfer, incompressible fluid dynamics, algorithms and data structures, ODE/PDE's, numerical analysis, intermediate solids and statics, aircraft performance and design
- Graduated as a Pratt Fellow for recognition of a 2+ year research project

EXPERIENCE

Atar Energy Solutions [thinktank]

Aug 2024 – Present

Co-Founder

Alexandria, VA

- Worked with Dr. Siavash Gholamhosseinpour to design, prototype, and build a novel FDS, Insulation Resistance and HiPOT Integrated Tester. It costs and weighs $\frac{1}{3}$ of the closest competition in a \$700M market
- Consulted with a handful of IP-law firms and negotiated deals with several potential private investors. Created the plan for non-provisional patent acquisition and legal entity creation
- Helped edit a 120-page technical datasheet and a 70-page professional plan

Edsall Auto Services

January 2025 – Present

Car Mechanic - Part Time

Alexandria, VA

- Go into the shop 2x a week to replace oil filters, spark plugs, brake pads, starter motors, filters, and certain sensors.
- Experience with 30+ makes/models from 1995 to 2023.

Duke General Robotics Lab

Aug 2022 – May 2024

Individual Research

Durham, NC

- Worked under Dr. Boyuan Chen to independently build and develop a versatile and accurate parallel-processed wave propagation simulation. This could emulate any material and lattice, any 3D shape, and input forces. Used to generate artificial data to train a custom A.I. that can "see with sound"
- Created a novel Chladni plate comparison test to connect simulated data to real-world vibration data. Achieved 90% accuracy. See website portfolio for the white paper and results
- Research approved by Duke Engineering Board for *Graduation with Distinction award*

FPrin LLC

Oct 2020 – Feb 2021

Lab Technician

Mountain View, CA

- Designed, built, and operated case-specific test devices to effectively characterize a recently patented German compressed gas-powered insulin pen. After that, stress tested thousands of components and maintained the shop/lab
- Designed and manufactured a functional self-balancing RC PID controlled robot named Balanciaga as an internal project

PROJECTS

- Senior Capstone Team Lead - designed, modeled, and tested a high-ISP electro-resistive hydrogen thruster for space application (300+ seconds ISP). Application sent to NASA for funding.
- Robot Design Studio - designed, built, programmed, and stress-tested a 1-foot bipedal walking and dancing robot
- Duke 24-hour Design Sprint Winner
- California Mayor Medal for A.I. forest fires study - 85% accuracy.
- FRC Robotics Quarter-finalist
- 2x Synopsys Science Fair Finalist (Engineering and CS)